

Introduction to NLP fundamentals

Prof. Luca Cagliero
Dipartimento di Automatica e Informatica
Politecnico di Torino



**Politecnico
di Torino**

Lecture goal

- Introduction to Natural Language Processing
 - overview of the main objectives, techniques, and application scenarios
- Describe traditional NLP fundamentals
 - The NLP pipeline, text preprocessing

Natural Language Processing

Default (GPT-3.5)

L

Can you define the Natural Language Processing?



Certainly! Natural Language Processing (NLP) is a subfield of artificial intelligence (AI) that focuses on the interaction between computers and human language. Its primary goal is to enable computers to understand, interpret, and generate human language in a way that is both meaningful and useful.

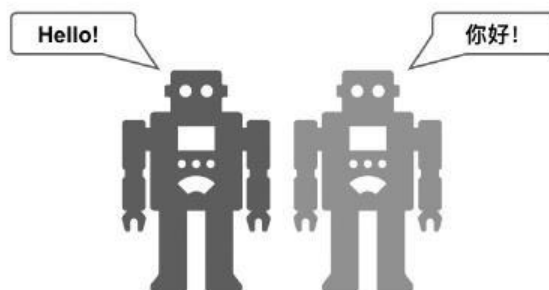
<https://chat.openai.com/>

Natural Language Processing

- Natural Language Processing (NLP) aims at enabling computers to process human languages intelligently
- Tight connection between several fields related to computer science and linguistics
 - Artificial Intelligence
 - Machine Learning
 - Cognitive science
 - Information processing
 - ...

Natural Language Processing

- NLP studies the connection between machine perception and languages
- Programming computers to fruitfully process large natural language corpora
- Examples of applications
 - Language understanding
 - Machine translation
 - Text generation
 - ...

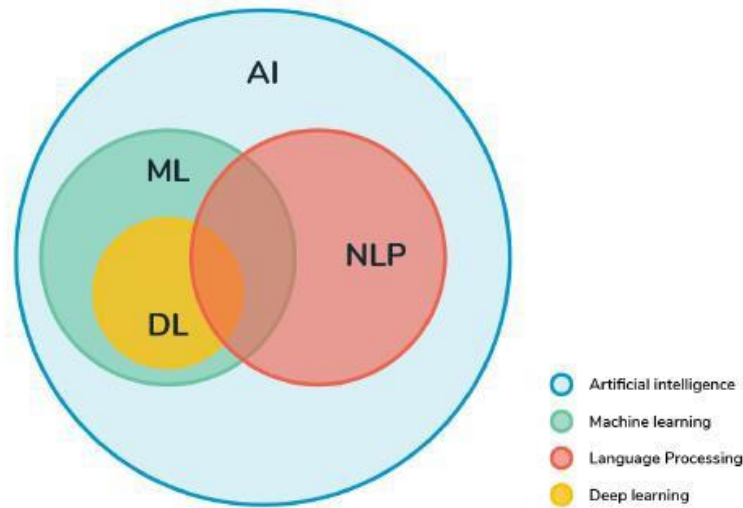


Traditional vs. Deep Natural Language Processing

- Traditional NLP techniques heavily relied on ad hoc linguistic models, e.g.,
 - Syntactic rules
 - Dictionaries
 - Text generation
 - ...
- NLP does not necessarily rely on automated learning

Traditional vs. Deep Natural Language Processing

- Deep NLP techniques leverage Machine Learning models to automate the learning process



Related buzzword: text mining

- The process of deriving significant information from text
 - Focus on unstructured and semi-structured text



Related buzzword: text mining

- The process of deriving significant information from text
- It encompasses
 - Information Retrieval (IR)
 - Machine Learning (ML)
 - Natural Language Processing (NLP)
 - Knowledge management (KM)

Related buzzword: Natural Language Understanding

- Natural Language Understanding or Interpretation (NLU/NLI)
 - NLP subtopic that deals with machine reading comprehension using AI techniques
 - It encompasses
 - Text categorization
 - Entity Recognition and Disambiguation
 - ...

Related buzzword: Natural Language Understanding

- Natural Language Generation (NLG)
 - It aims at generating human-like text or speech based on structured data or specific instructions
 - It encompasses
 - Summarization
 - Machine translation
 - Question Answering
 - ...

Overview of NLP applications

- NLP techniques find application in a wide range of knowledge discovery and decision support systems
- They are commonly part of a **data science pipeline**



Overview of NLP applications

- Hereafter we will focus on
 - Sentiment analysis
 - Text categorization
 - Machine translation
 - Question answering
 - Topic modelling
 - Text summarization

Sentiment analysis

- Extract writer's feeling, opinion, emotions, likes/dislikes
 - Also known as **opinion mining**
- Identify the opinion/human behavior of a person from plain text
- It can rely either on traditional NLP rules or on Machine Learning

"I am happy with this water bottle."



"This is a bad investment."



"I am going to walk today."



Sentiment analysis

- Use cases
 - Hotel review analysis
 - News trading
 - Hate speech detection
 - Advertisement placing
 - ...

Example of sentiment analyzer: VADER

- Input
`sentiment_analyzer_scores("The phone is super cool.")`
- Output
The phone is super cool --- {'neg': 0.0, 'neu': 0.326, 'pos': 0.674, 'compound': 0.7351}

Text categorization

- Assign a predefined label to a document or a text snippet
 - Also known as text classification
- Supervised process
 - given a set of labeled documents/text snippets, learn how to automatically label new, unlabeled ones
- If multiple labels are allowed
 - multi-label text classification
- It can rely either on traditional NLP rules or on Machine Learning

Text categorization

- Use cases
 - Spam detection
 - Ticket management
 - Automatic text annotation
 - Document management
 - ...



Machine translation

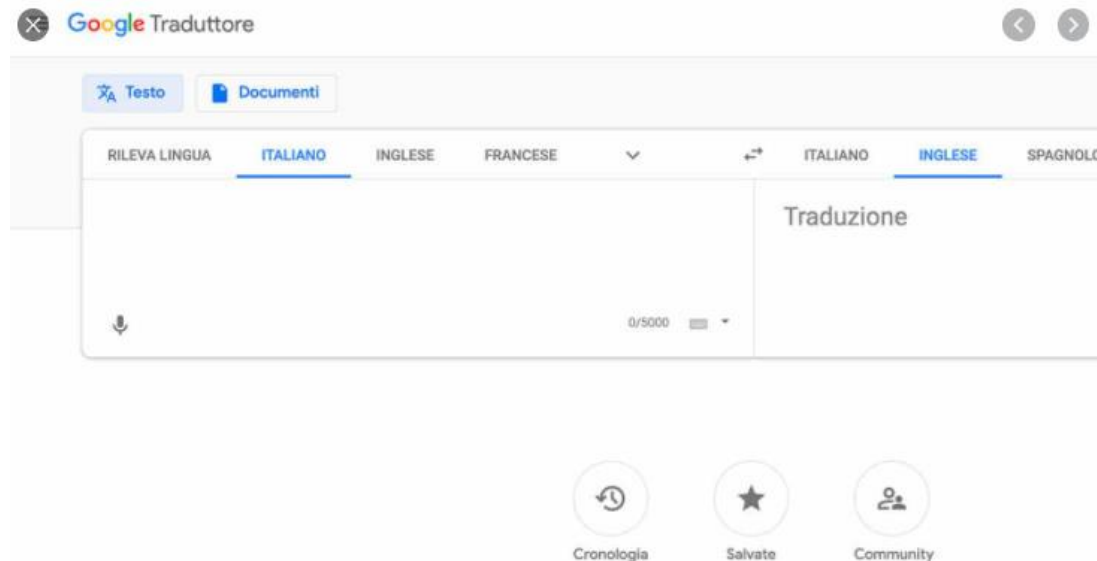
- Automatic translation of text or speech from one language to another
- Approaches
 - Rule- or dictionary-based
 - Statistical methods
 - Deep Learning



[memoq](#) (latest access: April 2021)

Rule-based Machine Translation

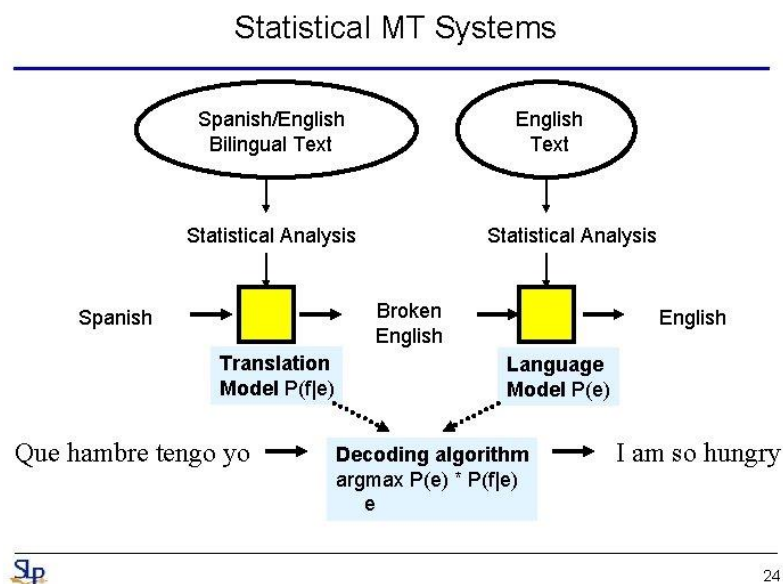
- Machine translation systems based on linguistic information about source and target languages basically retrieved from (bilingual) dictionaries and grammars
- Also known as **Knowledge-Based Machine Translation** or **Classical MT Approach**



<https://translate.google.com> (latest access: April 2021)

Statistical Machine Translation

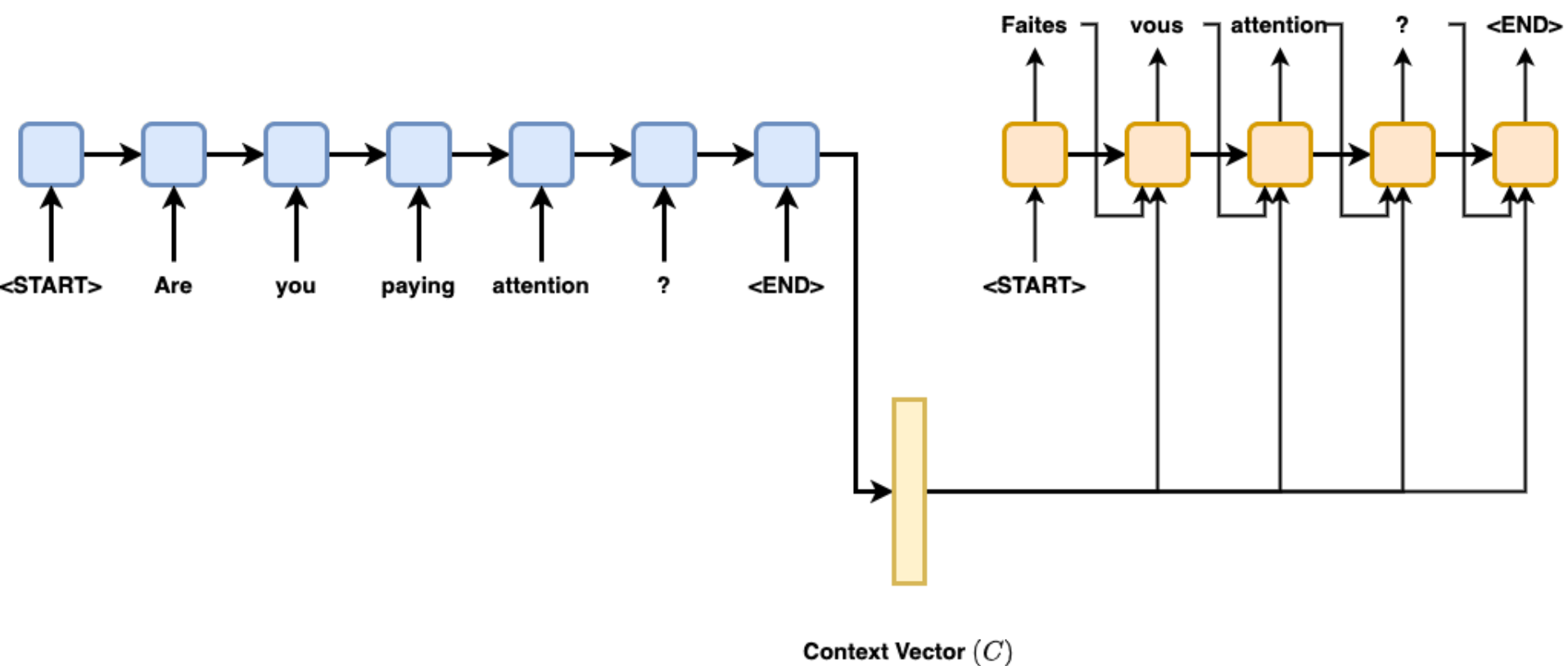
- Based on statistical models whose parameters are derived from the analysis of bilingual text corpora.
- Key ideas
 - every sentence in one language is a possible translation of any sentence in the other
 - the most appropriate is the translation that is assigned the highest probability by the system



Neural Machine Translation

ENCODER

DECODER



Neural Machine Translation

- The encoder in NMT creates a fixed-size context vector from all the hidden states of the encoder.
- The context vector is used by the decoder to get to the target sequence
- The decoder predicts the next word given the context vector and all the previously predicted words

Question answering

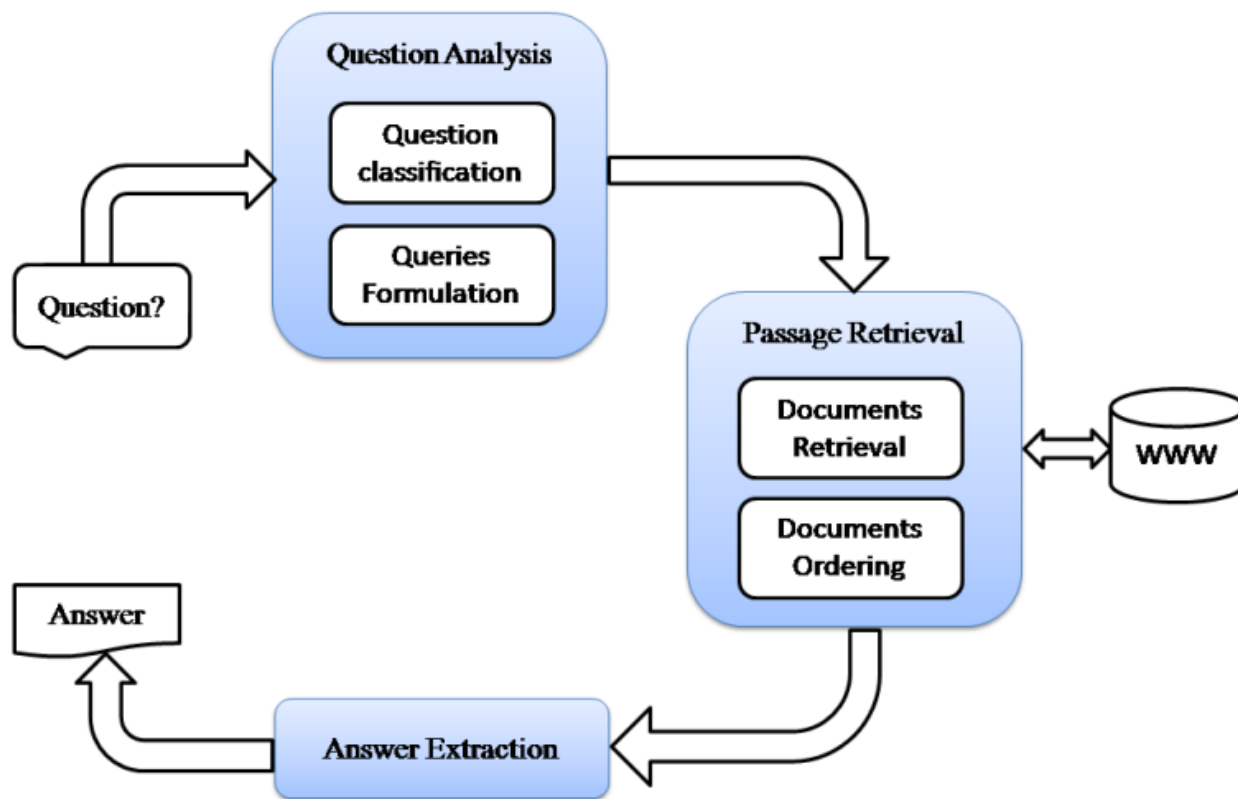
- Specialized area in the field of Information Retrieval
- It aims at providing relevant answers in response to questions posed in natural language
- Main steps
 - Question classification
 - Information Retrieval
 - Answer extraction



Question answering

- Key properties
 - Source: Web pages, knowledge bases, social data collections, etc.
- Question formulation
 - domain-specific, factoid, etc.
- Answer type
 - word, sentence, paragraph, image, video, etc.
- Practical use cases
 - Chatbots
 - Remote medical assistance
 - ...

Open-Domain Question Answering

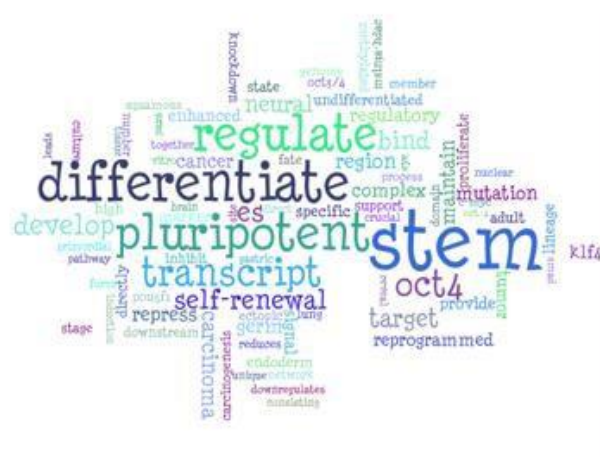


Conversational AI

- QA extension/adaptation
- Does not treat each question separately
 - Considers the history of past interactions and the relative context
 - Preserves the hierarchy and order of events

Topic modelling

- Unsupervised machine learning techniques aimed at detecting the key word and phrase patterns within a document collection
- A topic model is a statistical document representation of the abstract word groups and similar expressions that best characterize a set of documents



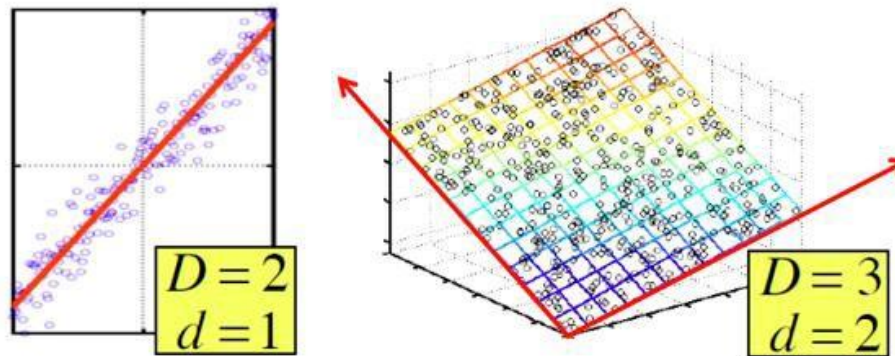
Topic modelling

- Main techniques
 - Latent Semantic Indexing (LSI)
 - Latent Dirichlet Allocation (LDA)
 - Aspect modelling
- Practical use cases
 - Information retrieval
 - Data visualization
 - ...

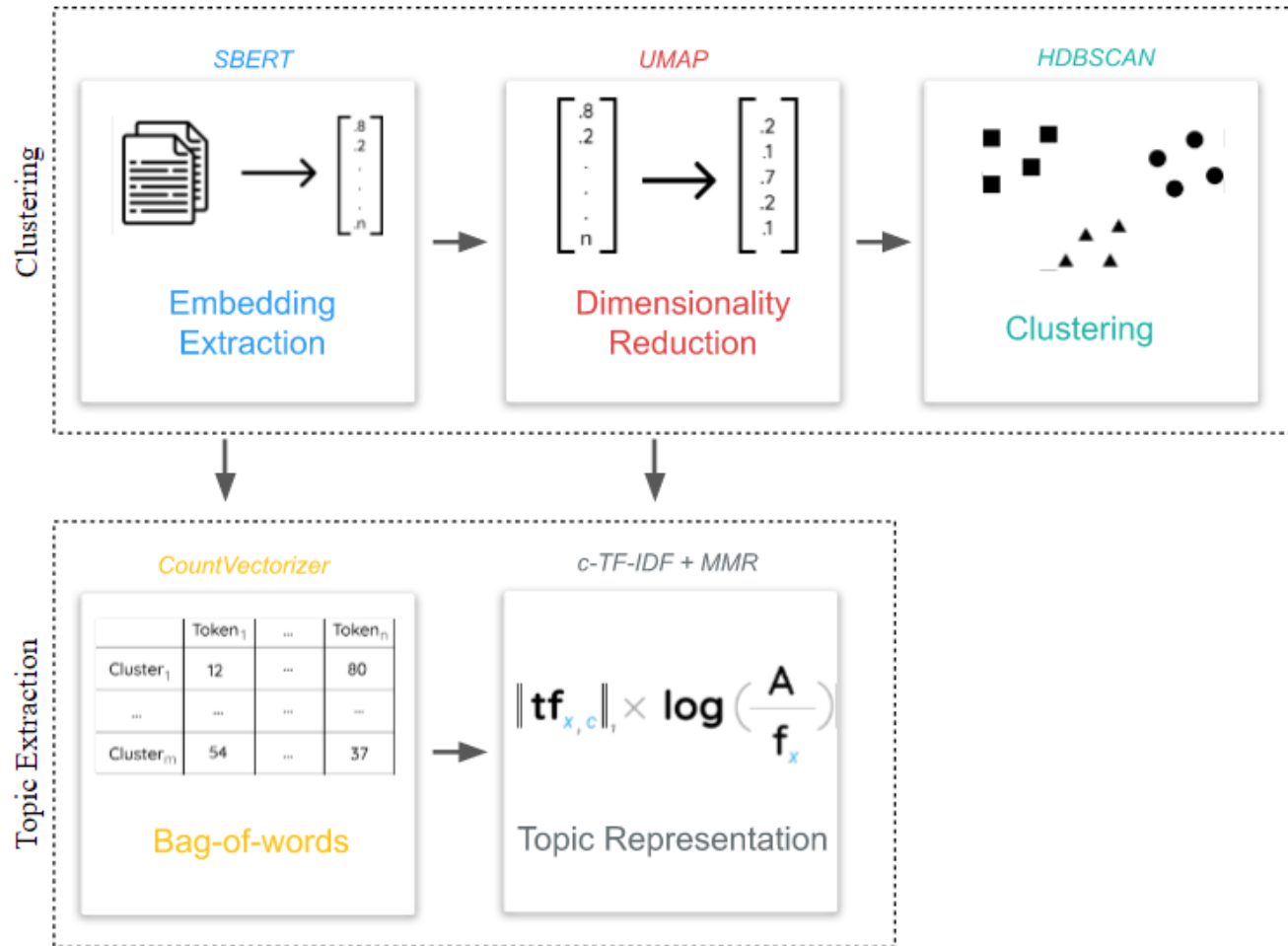
Topic modelling

- Dimensionality reduction

- Data lies on or near a low d -dimensional subspace
- Axes of this subspace are effective representation of the data



Topic Modeling - BERTopic



<https://github.com/MaartenGr/BERTopic>

Text summarization

- Shorten large textual document collections
 - Produce a concise summary that incorporates the most salient content

Transcript

today we are going to show you how to make spanish omelet . i 'm going to dice a little bit of peppers here . i 'm not going to use a lot , i 'm going to use very very little . a little bit more then this maybe . you can use red peppers if you like to get a little bit color in your omelet . some people do and some people do n't t is the way they make there spanish omelets that is what she says . i loved it , it actually tasted really good . you are going to take the onion also and dice it really small . you do n't want big chunks of onion in there cause it is just pops out of the omelet . so we are going to dice the up also very very small . so we have small pieces of onions and peppers ready to go .



Summary

how to cut peppers to make a spanish omelette; get expert tips and advice on making cuban breakfast recipes in this free cooking video .

Text summarization

- Input
 - Single document
 - Multi-document collection
 - Multimodal data
- Output
 - Extractive summary
 - pick existing content
 - Abstractive summary
 - generate new content
- Approach
 - Supervised
 - e.g., neural summarization
 - Unsupervised
 - e.g., itemset mining, clustering, Latent Semantic Analysis, graph ranking

Text summarization

- Language
 - Single-language
 - Multi-lingual
 - Cross-lingual
- Domain specificity
 - General-purpose
 - Context-dependent
 - Query-driven
- Time dependency
 - Temporal summarization
 - Timeline summarization
 - Incremental summarization

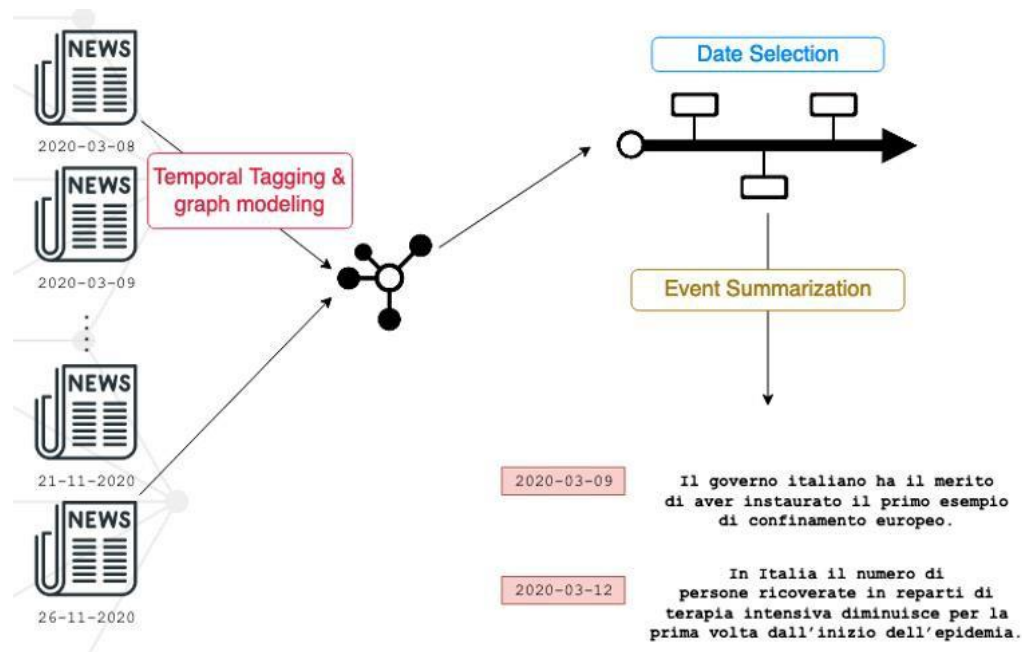
Text summarization

- Practical use cases
 - Content curation
 - Accessibility
 - Learning analytics
 - Highlight extraction



Examples of summarization methods

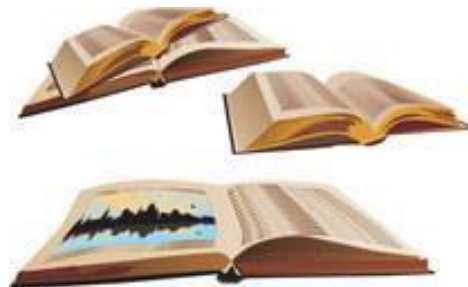
- TimeLine Summarization of news articles



Summarize Dates First: a paradigm shift in timeline summarization. Moreno La Quatra, Luca Cagliero, Elena Baralis, Maurizio Montagnulo, Alberto Messina.
ACM SIGIR 2021

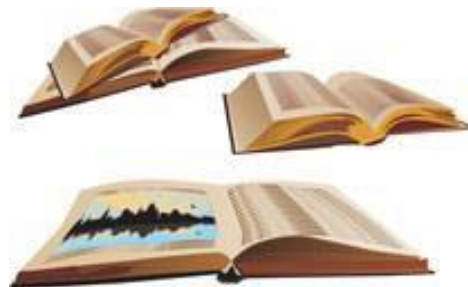
Text structure

- A text can be divided into units
 - Useful for enabling automated text processing
 - Crucial for extracting relevant knowledge
- Different aggregation levels
 - According to the objective
 - According to the language



Text structure

- Character: smallest text unit
 - Letter, digit, space, special character
 - Low utility unless enriched with positional information
 - Two words are commonly separated by a space
 - A full stop indicates the end of a sentence
 - A question mark indicates an interrogative form



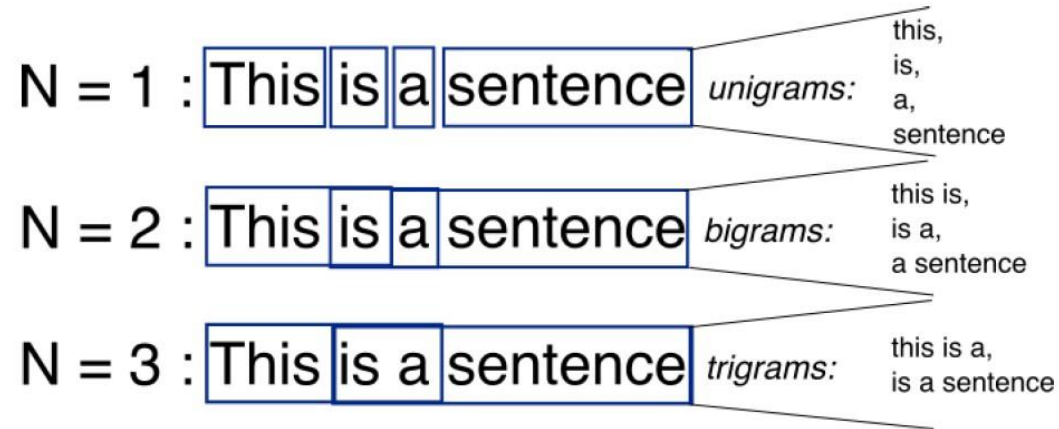
Text structure

- Word
 - series of letters between spaces
 - smallest unit of language that conveys meaning to the majority of people you would like to reach
 - Readers must agree with the language
- N-gram
 - contiguous sequence of N textual units
 - Units can be phonemes, syllables, letters, words
 - Syntactic and semantic relevance are not guaranteed
 - Unigram: 1-gram, bigram: 2-gram, etc.
- Multi-word expressions
 - textual form made up of at least two lexical items

Text structure

- Sentence
 - Text snippet separated by punctuation
 - e.g., full stop, question mark, exclamation mark
- Phrase
 - Part of a sentence consisting of a group of words without subject and verb
 - E.g., the temporal phrase after dinner or the impersonal form waiting for the bus
 - They do not convey a complete thought
- Clause
 - Part of a sentence including a subject and a predicate
 - It can be used by itself as a complete sentence
 - e.g., the independent clause the dog barked at him
 - It provides more insights (e.g., the subordinate clause when the old man came)

Text structure: examples of n-grams

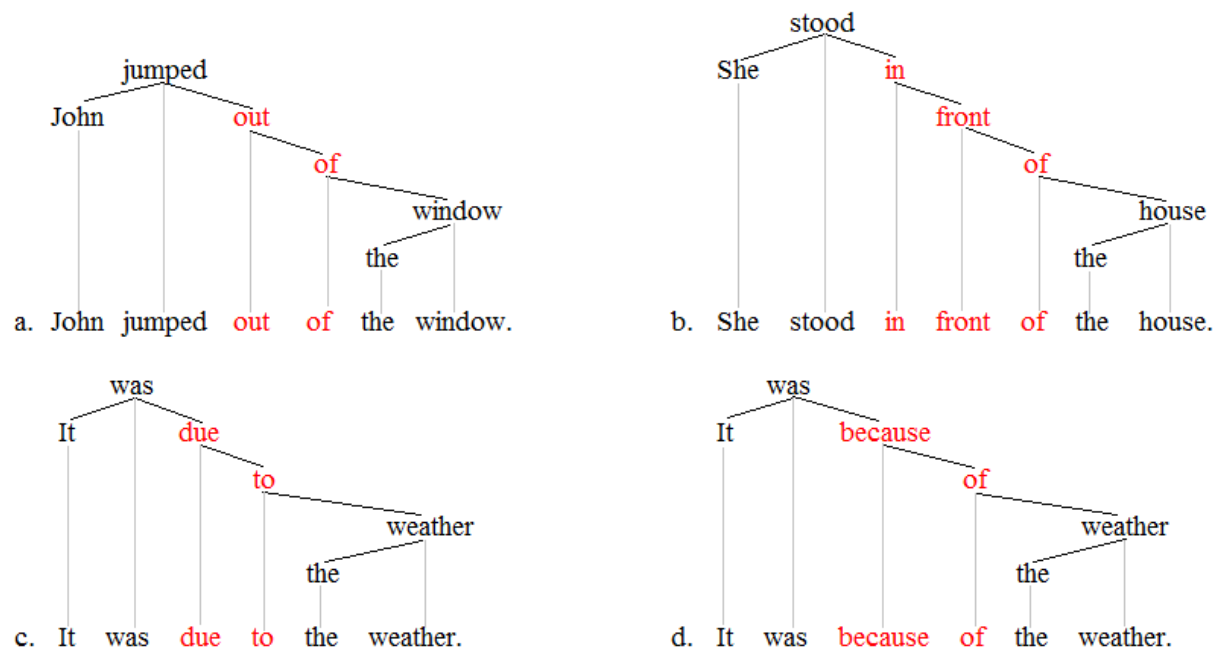


Source: <https://deeptai.org/> (latest access: April 2021)

Text structure

- Lexical chain
 - sequence of related words in writing
 - spanning short (adjacent words or sentences) or long distances (entire text)

Text structure: examples of lexical chains

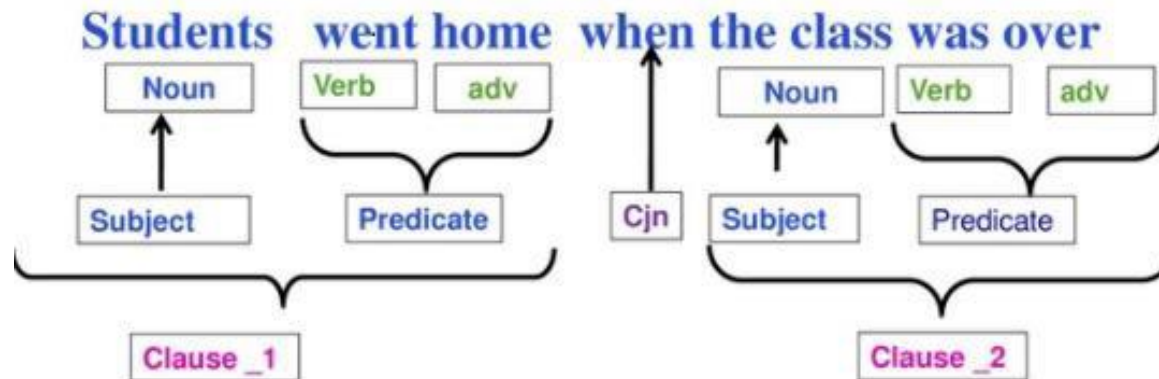


Source: <https://Wikipedia.org> (latest access: April 2021)

Text structure

- **Lemma**
 - canonical form of a word or a multi-word expression, chosen from a set of candidate forms in a dictionary
- **Lexeme**
 - set of all single words or multi-word forms that have the same meaning
 - A lemma is a particular form that is chosen by convention to represent the lexeme
 - E.g., the same word with different inflectional endings
- **Stem**
 - base form of a word (similar to a lemma) not necessarily derived from a dictionary
 - Typically, static rules to transform inflected forms
- **Bag-Of-Word (BOW)**
 - unordered set of words
 - Repetitions are removed

Text structure: examples of clauses and phrases

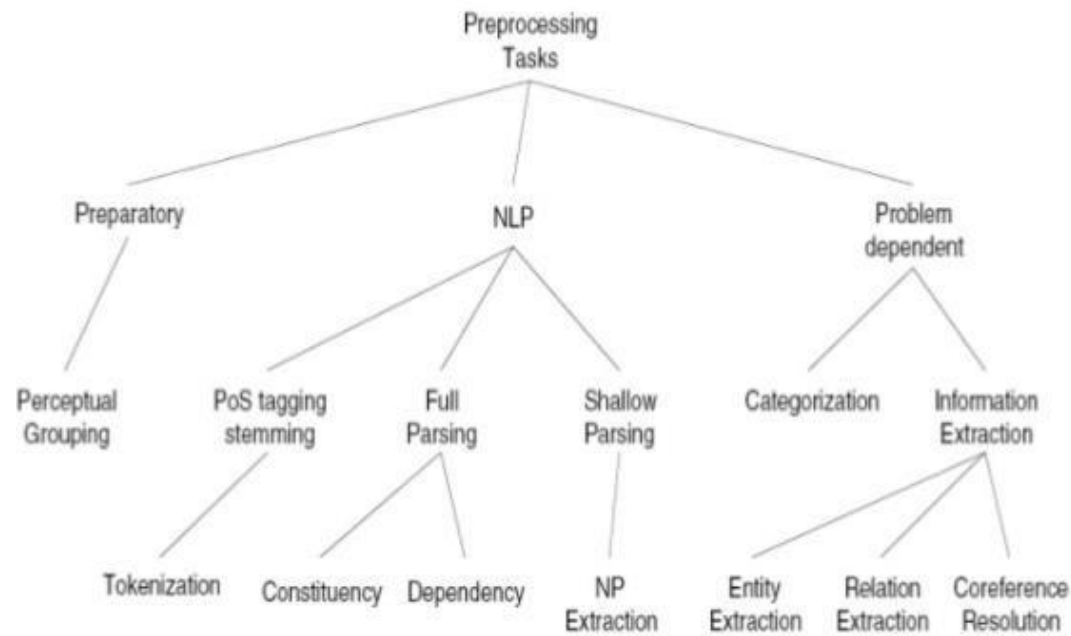


Source: <https://knowitinfo.com/> (latest access: April 2021)

Text structure: sectioning

- Paragraph
 - Portion of text consisting of a sequence of sentences
 - Paragraphs can be further partitioned into subparagraphs
- Section
 - Portion of text consisting of consecutive paragraphs
 - Sections can be further partitioned into subsections and subsubsections

Text preprocessing: the taxonomy



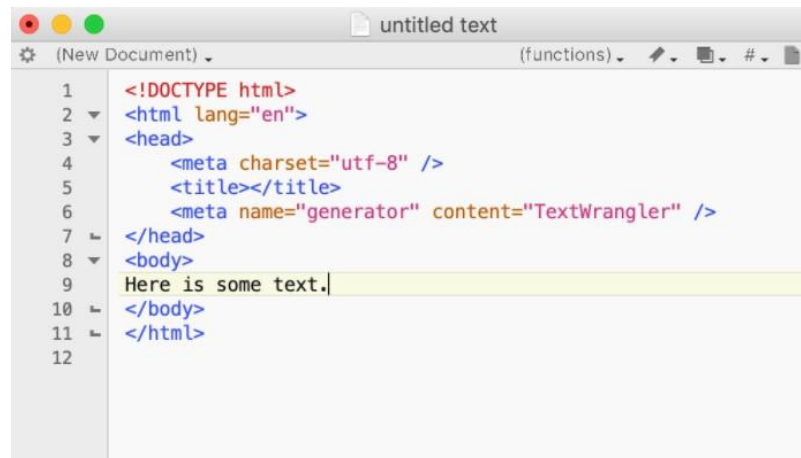
The Text Mining Handbook Advanced Approaches in Analyzing Unstructured Data. Ronen Feldman, James Sanger, 2007. ISBN: 9780521836579

Text preprocessing steps

- Cleaning
 - basic filtering steps to remove noise, errors, and redundant content
- Tokenization
 - divide the raw text into units and sub-units
- Stopword elimination
 - remove too frequent words with little semantic meaning
- Part-Of-Speech tagging
 - annotate the text words with the corresponding role in the sentence
- Lemmatization and stemming
 - map word inflections and derivatives to their canonical form

Text cleaning

- Remove special characters and extra spaces
 - unrecognized symbols, OCR errors, etc.
- Case normalization
 - handle upper and lower cases
- Data format conversion
 - E.g., HTML, XML, JSON, CSV
 - May require a data schema
- Handle non-textual content



The screenshot shows a code editor window titled "untitled text". The editor displays an HTML document structure with the following code:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="utf-8" />
5   <title></title>
6   <meta name="generator" content="TextWrangler" />
7 </head>
8 <body>
9   Here is some text.
10 </body>
11 </html>
12
```

Language dependence

- Cleaning ✓ (partially)
- Tokenization ✓
- Stopword elimination ✓
- Part-Of-Speech tagging ✓
- Lemmatization and stemming ✓

**What if you do not know
the source language in advance?**

Tokenization

- Continuous character stream in documents is broken up into meaningful constituents
- Break text into relevant units and sub-units
- Identify sentence boundaries based on punctuation, capitalization, etc.
- Strongly language-dependent



Source: <https://kddnuggets.com> (latest access: April 2021)

Example of tokenizer

Text tokenization utility

Source: `R/preprocessing.R`

Vectorize a text corpus, by turning each text into either a sequence of integers (each integer being the index of a token in a dictionary) or into a vector where the coefficient for each token could be binary, based on word count, based on tf-idf...

```
text_tokenizer(  
  num_words = NULL,  
  filters = "!\"#$%&()*+,-./:;<=>?@[\\]^_`{|}~\t\n",  
  lower = TRUE,  
  split = " ",  
  char_level = FALSE,  
  oov_token = NULL  
)
```

Arguments

- num_words** the maximum number of words to keep, based on word frequency. Only the most common `num_words` words will be kept.
- filters** a string where each element is a character that will be filtered from the texts. The default is all punctuation, plus tabs and line breaks, minus the ' character.
- lower** boolean. Whether to convert the texts to lowercase.
- split** character or string to use for token splitting.
- char_level** if `TRUE`, every character will be treated as a token
- oov_token** `NULL` or string. If given, it will be added to ``word_index`` and used to replace out-of-vocabulary words during `text_to_sequence` calls.

Source: https://keras.rstudio.com/reference/text_tokenizer.html (latest access: April 2021)

Stopword elimination

- Stop words refer to the most common words in a language
- They convey a limited amount of information
 - E.g., prepositions, articles, conjunctions
- They are filtered out prior to text processing

anything that has mass and
occupies space is called matter.
matter exists in various states
such as

Source: <https://kdnuggets.com> (latest access: April 2021)

Stopword elimination

- Language-dependent step
- Some particular Deep NLP models do not require stopwords elimination
- A universal list of stop words used by all natural language processing tools is not available
- Ideally, any group of words can be chosen as the stopwords for a given purpose
 - Different search engines use different stopwords lists
 - Some of them remove lexical words, such as want, from a query in order to improve performance

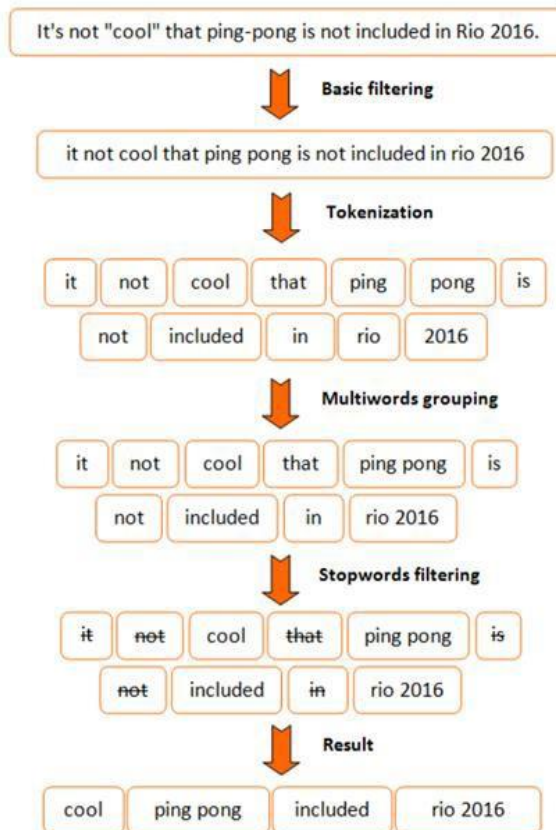
Example of stopwords list

```
NLTK's list of english stopwords

1  i
2  me
3  my
4  myself
5  we
6  our
7  ours
8  ourselves
9  you
10 your
11 yours
12 yourself
13 yourselves
14 he
15 him
16 his
17 himself
18 she
19 her
20 hers
21 herself
22 it
23 its
24 itself
25 they
26 them
27 their
28 theirs
29 themselves
30 what
31 which
32 who
```

Source: <https://gist.github.com/sebleier/554280> (latest access: April 2021)

Example of text preprocessing pipeline



Source: www.meaningcloud.com (latest access: April 2021)

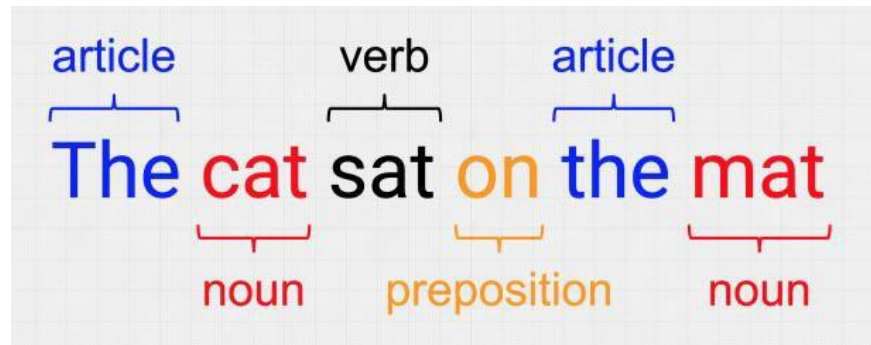
Example of text preprocessing pipeline

- Caveat
 - By removing stopword *Not* the meaning of the text changes!
- To capture the semantics behind the text, stopword elimination is deprecated
 - Deep NLP techniques are recommended

Part-Of-Speech tagging

- Part-Of-Speech (POS)
 - word category assigned according to the role it plays in the sentence
- POS tagging
 - label the words in a text with the corresponding POS
 - E.g., article, noun, verb, adjective, preposition, number, and proper noun
- Context and language dependent
- Relies on morphological word analyses

Part-Of-Speech tagging example



Source: <https://dataingovernment.blog.gov.uk/> (latest access: April 2021)

Lemmatization

- Process of determining the lemma of a word by deducing its intended meaning
 - It groups together the inflected forms of a word
 - They can be considered as the same thing and identified by a common lemma
- It can perform together with POS tagging
- It may require analyzing the sentence- or document-level context
 - E.g., article, noun, verb, adjective, preposition, number, and proper noun
- Context and language dependent
- Relies on morphological word analyses

Stemming

- Simplified version of the lemmatization process
- It determines the stem of the word instead of the lemma
- Lower computational complexity
- Approximated result
- Acceptable in most cases

Form	Suffix	Stem
studies ^{es}	-es	studi
study ^{ing}	-ing	study
niñ ^{as}	-as	niñ
niñ ^{ez}	-ez	niñ



Source: <https://devopedia.org/> (latest access: April 2021)

Stemming vs. lemmatization

Stemming vs Lemmatization

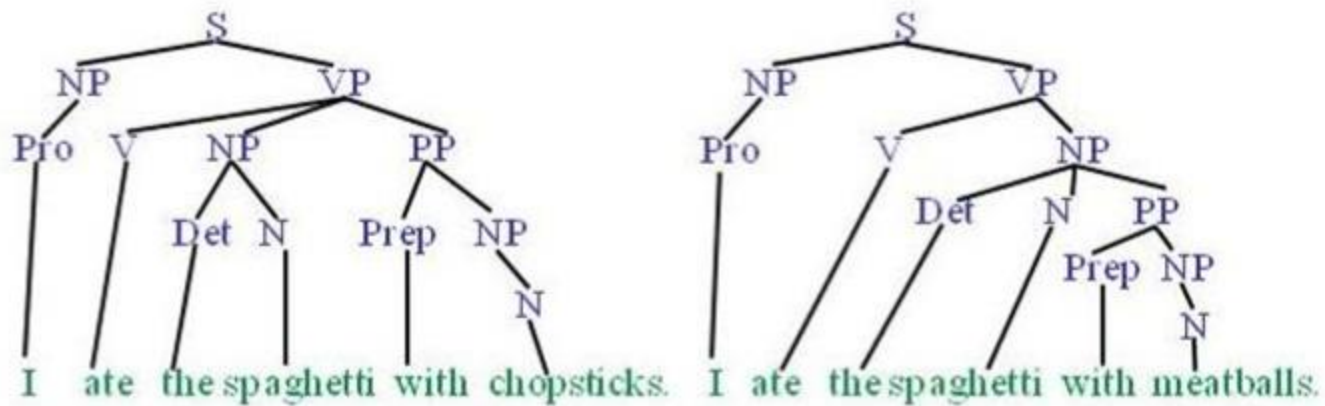


Source: <https://dataingovernment.blog.gov.uk/> (latest access: April 2021)

Syntactic parsing

- Syntactical analysis of the text according to a certain grammar theory
- Constituency grammars describe the syntactical structure of the sentence in terms of recursively built phrases,
 - i.e., sequences of syntactically grouped elements
- Dependency grammars analyze the dependencies between words
 - E.g., an adjective depends on the associated noun
- High computational complexity
- Low quality in many real contexts

Syntactic parsing

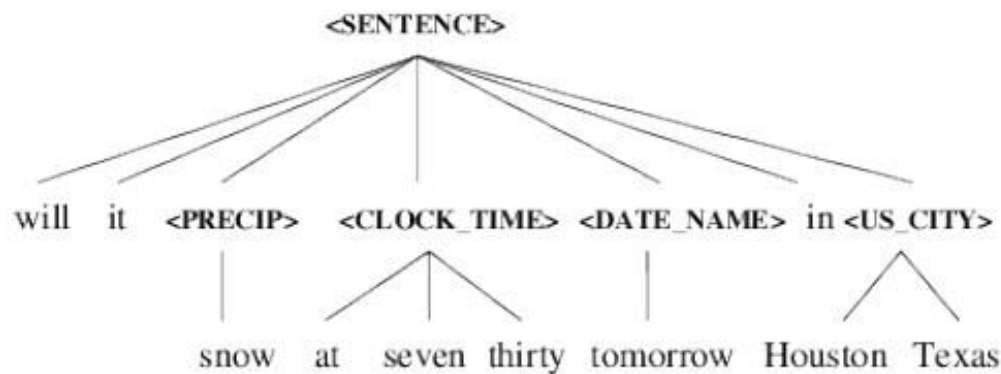


Parsing example: “book that flight”

$S \rightarrow NP VP$	$Det \rightarrow that \mid this \mid a$
$S \rightarrow Aux NP VP$	$Noun \rightarrow book \mid flight \mid meal \mid money$
$S \rightarrow VP$	$Verb \rightarrow book \mid include \mid prefer$
$NP \rightarrow Pronoun$	$Pronoun \rightarrow I \mid she \mid me$
$NP \rightarrow Proper-Noun$	$Proper-Noun \rightarrow Houston \mid TWA$
$NP \rightarrow Det Nominal$	$Aux \rightarrow does$
$Nominal \rightarrow Noun$	$Preposition \rightarrow from \mid to \mid on \mid near \mid through$
$Nominal \rightarrow Nominal Noun$	
$Nominal \rightarrow Nominal PP$	
$VP \rightarrow Verb$	
$VP \rightarrow Verb NP$	
$VP \rightarrow Verb NP PP$	
$VP \rightarrow Verb PP$	
$VP \rightarrow VP PP$	
$PP \rightarrow Preposition NP$	

Shallow parsing

- Simplified text parsing
- Syntactical analysis only of the text snippets that are unambiguous
 - E.g., small and simple noun and verb phrases
- Trade-off between complexity and accuracy



Named Entity Recognition

- Locate and classify names in text
- Identify references to information units called entities
- Entities are semantically rich (ontology-based) descriptions of concepts/bodies
 - E.g., persons, organization and location names, numeric
 - expressions including time, date, money, and percent expressions, times, dates, proteins, etc.

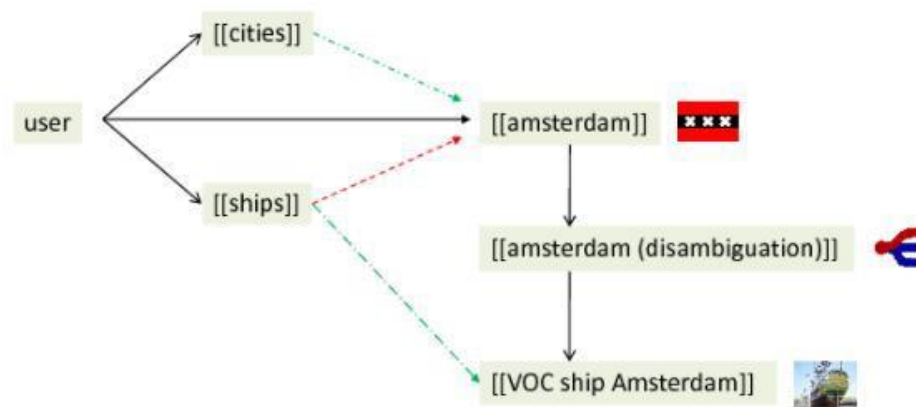
Named Entity Recognition

contentSkip to site indexPoliticsSubscribeLog InSubscribeLog InToday's PaperAdvertisementSupported **ORG** byF.B.I. Agent Peter Strzok **PERSON** ,
Who Criticized Trump **PERSON** in Texts, Is FiredImagePeter Strzok, a top F.B.I. **GPE** counterintelligence agent who was taken off the special counsel
investigation after his disparaging texts about President Trump **PERSON** were uncovered, was fired. CreditT.J. Kirkpatrick **PERSON** for The New York
TimesBy Adam Goldman **ORG** and Michael S. SchmidtAug **PERSON** . 13 **CARDINAL** , 2018WASHINGTON **CARDINAL** — Peter Strzok
PERSON , the F.B.I. **GPE** senior counterintelligence agent who disparaged President Trump **PERSON** in inflammatory text messages and helped
oversee the Hillary Clinton **PERSON** email and Russia **GPE** investigations, has been fired for violating bureau policies, Mr. Strzok **PERSON** 's lawyer
said Monday **DATE** .Mr. Trump and his allies seized on the texts — exchanged during the 2016 **DATE** campaign with a former F.B.I. **GPE** lawyer,
Lisa Page — in **PERSON** assailing the Russia **GPE** investigation as an illegitimate “witch hunt.” Mr. Strzok **PERSON** , who rose over 20 years
DATE at the F.B.I. **GPE** to become one of its most experienced counterintelligence agents, was a key figure in the early months **DATE** of the
inquiry.Along with writing the texts, Mr. Strzok **PERSON** was accused of sending a highly sensitive search warrant to his personal email account.The
F.B.I. **GPE** had been under immense political pressure by Mr. Trump **PERSON** to dismiss Mr. Strzok **PERSON** , who was removed last summer
DATE from the staff of the special counsel, Robert S. Mueller III **PERSON** . The president has repeatedly denounced Mr. Strzok **PERSON** in posts on

Named Entity Recognition and the Stanford NER Software. Jenny Rose. Stanford University. 2007 27

Word sense disambiguation

- Identifying which sense of a word is used in a sentence, when the word has multiple meanings
- Approaches
 - knowledge base exploration
 - Machine Learning techniques
 - a combination of the above
- Acronym NERD stands for "Named Entity Recognition and Disambiguation"



Word disambiguation example: “cars”

- Same Google query, different results



Text representation

- Transform the raw text into a data representation suitable for applying Machine Learning and data mining algorithms
 - Most algorithms require a structured data model (relational, key-value form, or equivalent)
 - A fixed-sized model is requested
- Main issue
 - Move from an unstructured or semi-structured text representation to a structured form

Text representation

- Fully unstructured document
 - Raw text without sectioning and paragraphs

Customer XYZ called about Plan A promotion. Explained plan.
Customer thinks roll-over minutes should be included.

Customer ABC called about Plan A promotion. Customer thought
it was ridiculous that roll-over minutes were not in plan.

Potential called about Plan A promotion. Said that plan was expensive.

Potential called about Plan A promotion. Said that 4GB data not enough.

Customer XYT called about Plan A promotion. Said
that data plan was insufficient and stupid.

Text representation

- Weakly structured document
 - Text organized in sections, paragraphs according to a predefined format

1 Section

Hello World!

1.1 Subsection

Structuring a document is easy!

1.1.1 Subsubsection

More text.

Paragraph Some more text.

Subparagraph Even more text.

2 Another section

Text representation

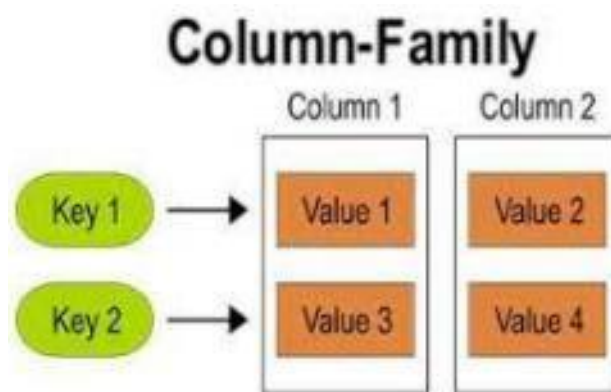
- Semi-structured document
 - Text annotated with tags or with a overlay structure based on markups
 - e.g., XML, HTML

```
<CATALOG>
  <SPRING>
    <TITLE>Garden Sales</TITLE>
    <LINE>Outdoor_Tools</LINE>
    <PAGE>
      <CAPTION>Goodbye, Winter!</CAPTION>
      <ITEM>Gardening Gloves</ITEM>
      <ITEM>Potting Soil</ITEM>
    </PAGE>
  </SPRING>
</CATALOG>
```

Feature-value data model

- Structured way to represent the main information
- A set of features summarizing the key properties of the text
 - E.g., the words that occur in it
- A set of textual units
 - E.g., the sentences

Text representation example: column-based DBs



Feature-value data model

- Structured way to represent the main information
- A set of features summarizing the key properties of the text
 - E.g., the words that occur in it
- A set of textual units
 - E.g., the sentences

Feature-value data model: main steps

- Feature engineering
 - define a set of features and collect the corresponding values
 - considering various textual units and heuristics based on the application domain
- Data transformation
 - build a structured representation of the original text that incorporates the considered features

Feature engineering

- Dimensionality reduction
 - Reduce the number of features to alleviate the issues related to high-dimensional data
 - Main methods
 - Latent Semantic Indexing, Latent Dirichlet Analysis, Principal Component Analysis
- Feature selection
 - Reduce the number of features to improve the performance of Machine Learning models
 - e.g., sentiment analysis models, text classifiers

Feature selection

- Unsupervised methods
 - Do not use the target variable (i.e., remove redundant variables)
 - E.g., Pearson correlation analysis
- Supervised methods
 - Use the target variable (e.g. remove irrelevant variables).
 - Wrapper
 - Search for well-performing subsets of features
 - Filter
 - Select subsets of features based on their relationship with the target
 - Statistical Methods
 - Feature Importance Methods
 - Intrinsic methods
 - Algorithms that perform automatic feature selection during training
 - E.g., Decision Trees

Acknowledgements and copyright license



- Copyright licence
 - Attribution + Noncommercial + NoDerivatives
- Affiliation
 - The author and his staff are currently members of the Database and Data Mining Group at Dipartimento di Automatica e Informatica (Politecnico di Torino) and of the SmartData interdepartmental centre
 - <https://dbdmg.polito.it>
 - <https://smartdata.polito.it>

Thank you!